



AI Real Estate Intelligence Platform

Predict prices · Discover explained matches · Ask an AI advisor · Chat with your property documents

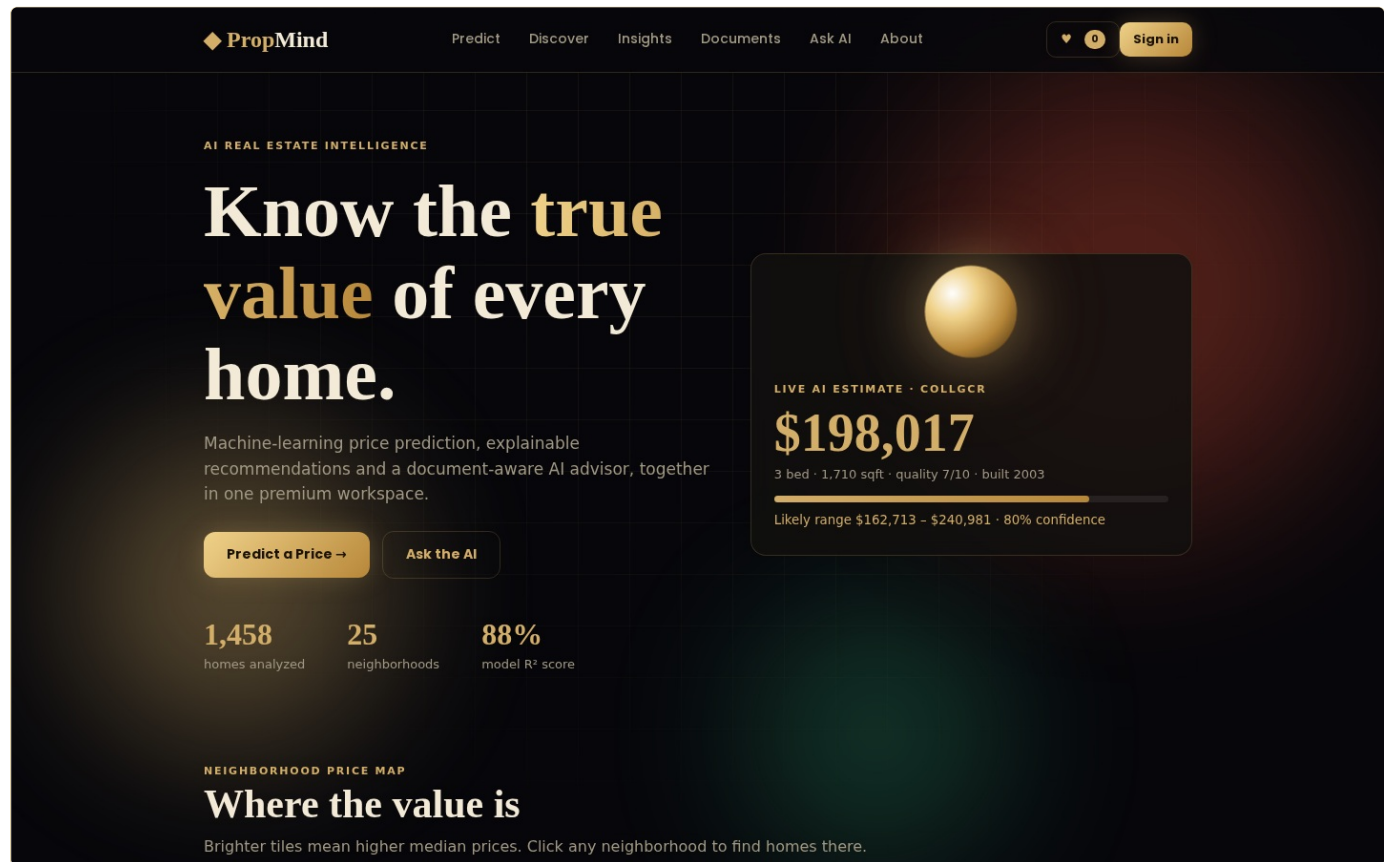


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About the project

Buying a home means answering three questions with very little reliable help: *What is this property really worth? Which homes fit me, and why? What do the legal and financial documents actually say?*

PropMind answers all three in one product. It combines data analysis, machine learning, generative AI and retrieval-augmented generation (RAG) behind a FastAPI backend and a premium dark-luxury web interface. It was built as the final project for the AI and Data Science course at SMIT (Saylani Mass IT Training), and is designed as a real-world AI product rather than a standalone model.

Key features

Feature	What you get
◆ AI price prediction	A Random Forest estimate from nine simple inputs, an 80% likely range, a confidence score and <i>what-if</i> effects (" +1 quality adds about \$X")
★ Explained recommendations	Ranked homes by budget, location, bedrooms and preferences. Every result lists why it was recommended and how its price compares with the AI fair value
✚ AI real estate advisor	A Groq-powered chatbot with prompt engineering, market context, guardrails and conversation memory
□ Document RAG assistant	Answers grounded in four real-world property documents (plus your own PDF/TXT/MD uploads) with [1] citations and source passages
▮ Market insights	Interactive neighborhood rankings, model comparison and eight analysis charts
♥ Shortlist, compare and report	Save favourites, compare up to three homes side by side, download a PDF valuation report, keep valuation and chat history
⚙ Production touches	Demo login, offline fallback when no API key is set, input validation, CORS, 10 automated tests, static build for Firebase

Product tour

Price prediction

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MACHINE LEARNING

Price Prediction

Describe a property and the trained Random Forest model estimates its fair market price instantly.

Neighborhood

NridgIt

Living area

1,700 sqft

Bedrooms

123456

Full bathrooms

1234

Overall quality

9/10 - Excellent

Year built

2003

Garage capacity (cars)

01234

Basement

650 sqft

ESTIMATED SALE PRICE

\$250,296

80% likely range: \$205,671 - \$304,603

PER SQFT

\$147

VS AREA MEDIAN

-20.5%

CONFIDENCE

80%

PDF reportFind similar homesSave valuation

Model: Random Forest - R² 0.8846. Educational estimate based on Ames, Iowa sales (2006-2020). Not an appraisal.

WHAT-IF ANALYSIS

What would change the price?

+1 overall quality +\$1,088

+200 sqft living area +\$6,418

+1 garage space +\$34,622

Smart recommendations

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RECOMMENDATION ENGINE

Discover Homes

Tell us your budget and preferences. Every match comes with an explanation of why it was recommended.

Budget

\$250,000

Minimum bedrooms

1+2+3+4+5+

Preferred neighborhoods

(1)BrimleyBlairstBrOde

BESideClearCrCollyCr

CrawforEdwardGilbert

IDOTRAHeadwayMichel

Preferences

2+ car garageNewer build (2000+)

High qualityLarge lotSpacious

Big basement

Find homes

Recommendations

▼ Saved (2)

108 homes fit - showing top 6

#1 MATCH - COLLOC

\$245,350

9.6% below AI value

3 bed2 bath1,694 sqftQuality 8/10Built 20083 car

Air fair value \$268,289 - \$145/sqft

✓ Priced at \$245,350, \$4,650 under your \$250,000 budget

✓ Listed 9% below the AI fair-value estimate of \$268,289

✓ 3 bedrooms, meeting your 3+ requirement

✓ Located in CollyCr, your preferred area

✓ Matches your preference - has a 3-car garage

This 3-bedroom home in CollyCr offers 1,694 sqft, was built in 2008 and has a quality rating of 8/10. It is priced at \$245,350, about 9% below the AI fair-value estimate of \$268,289. Verify condition with an inspection before making an offer.

or CompareAI summary

2 of 3 selectedCompareClear

Compare homes

Compare homes

COLLOC #1189COLLOC #1148

Price	\$245,350	\$228,500
AI fair value	\$268,289	\$253,115
Value vs AI	+9.6%	+9.7%
Neighborhood	CollyCr	CollyCr
Bedrooms	3	3
Bathrooms	2	2
Living area	1,694 sqft	2,389 sqft
Quality	8/10	9/10
Year built	2008	2002
Garage	3 car	2 car
Basement	1,084 sqft	1,284 sqft
Lot	10,478 sqft	9,375 sqft
Price per sqft	\$145	\$105

Green marks the best value in each row.

Market insights

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DATA ANALYSIS

Market Insights

Explore price patterns across neighborhoods, and see how the machine-learning models performed.

HOUSES ANALYZED

1,458

MEDIAN PRICE

\$163,000

NEIGHBORHOODS

25

MODEL ACCURACY (R²)

0.8846

LOCATION COMPARISON

Neighborhoods ranked

Median pricePrice per sqftQualityMedian year built

Intrepid	\$188	\$158	\$144	\$107
Stonewell	\$187	\$157	\$106	\$104
Wentker	\$187	\$157	\$106	\$103
Somerset	\$187	\$157	\$106	\$102
Brimley	\$187	\$157	\$106	\$101
CollyCr	\$187	\$157	\$106	\$100
Timber	\$187	\$157	\$106	\$99
Michel	\$187	\$157	\$106	\$98
Headway	\$187	\$157	\$106	\$97
Sageyer	\$187	\$157	\$106	\$96
Names	\$187	\$157	\$106	\$95
ClearCr	\$187	\$157	\$106	\$94
Gilbert	\$187	\$157	\$106	\$93
Sageyer	\$187	\$157	\$106	\$92
Crawfor	\$187	\$157	\$106	\$91

Document RAG assistant

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RETRIEVAL-AUGMENTED GENERATION

Document Knowledge Assistant

Ask questions about buying, legal steps, investing and neighborhoods. Answers come only from the indexed documents, with citations.

KNOWLEDGE BASE

Ames Area InformationIndexed / 2,361 words - 32 passages

Investment TipsIndexed / 879 words - 12 passages

Registry & Legal ProcessIndexed / 1,080 words - 12 passages

Property Buying GuideIndexed / 1,134 words - 13 passages

Drop a PDF, TXT or MD file or click to upload (max 3 MB)

Uploaded documents are indexed instantly and searchable in your questions.

ASK YOUR DOCUMENTS

What is the transfer tax in Iowa?

Ask

How much down payment does an FHA loan need?

What is the transfer tax in Iowa?

How do I calculate cap rate?

Which neighborhood is the cheapest?

What should I check during a home inspection?

How are property taxes paid in Iowa?

From Registry & Legal Process (6. Taxes and Fees at Closing) - Real estate transfer tax: Iowa charges a transfer tax calculated per \$500 of value, with the first \$500 exempt. Confirm the current rate with the county recorder. It is customarily paid by the seller. • Recording fees: paid to the county recorder for recording the deed and mortgage. • Property taxes in Iowa, property taxes are billed in two installments, due by the end of September and the end of March, and are... [1]

Offline mode: answering the best matching passage

SOURCES

[1] Registry & Legal Process (6. Taxes and Fees at Closing) (relevance 0.2341)

• Real estate transfer tax: Iowa charges a transfer tax calculated per \$500 of value.

AI advisor

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GENERATIVE AI

PropMind AI Advisor

Ask about prices, neighborhoods, market trends or buying guidance.

OFFLINE MODE

Clear chat

Hello! I'm PropMind, your AI real estate advisor. I can compare neighborhoods, explain prices and walk you through buying decisions. What would you like to know?

What is NridgIt like?

NridgIt has a median sale price of \$315,000 (about \$156 per sqft) in the Ames data. Use the Price Prediction page for a home-specific estimate. (Offline mode: add GROQ_API_KEY for full AI answers.)

Offline mode

Which neighborhoods are most affordable?

Is NridgIt worth the premium?

Explain the home buying process

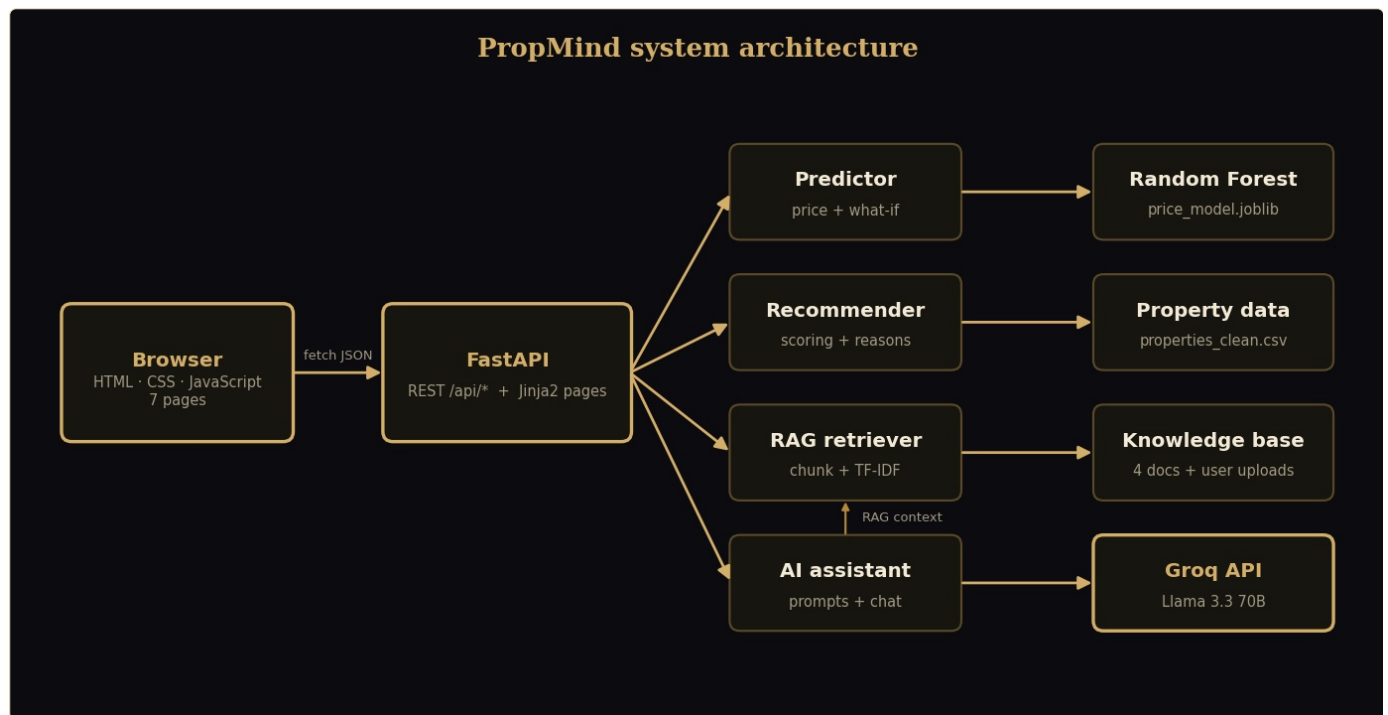
What makes a house price go up?

Ask anything about real estate

Send

Screenshots were captured without a Groq key, so the assistant shows its offline mode. With a key, answers are written by the LLM.

System architecture

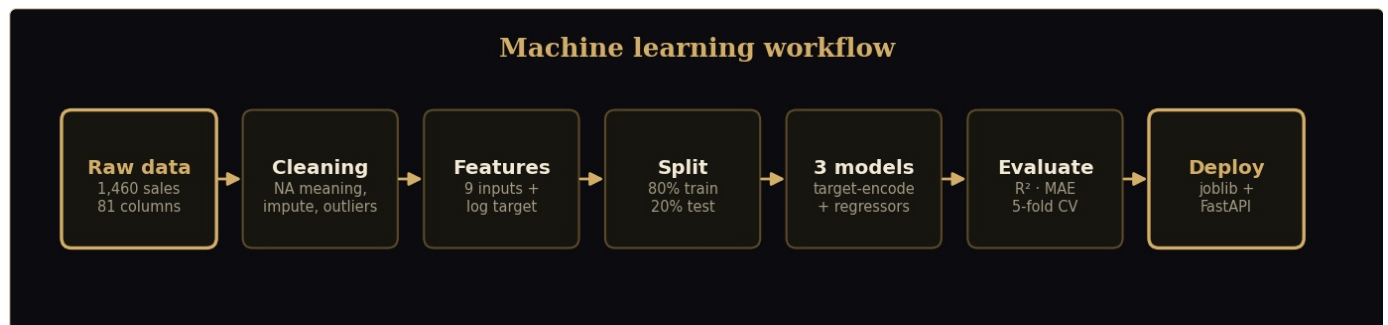


- The **browser** talks to FastAPI through a small JSON API (/api/*). The same FastAPI app also renders the seven Jinja2 pages.
- **Services** are plain Python modules (no web dependencies), so they are easy to test and reuse.
- The **assistant** builds prompts and calls the Groq API. If no key is configured, it degrades gracefully to an offline mode instead of failing.

How it works

1. Data analysis and preprocessing

The Kaggle *Ames Housing* data (1,460 sales, 79 features) is cleaned with Pandas and NumPy: "NA" values that mean *feature absent* (pool, alley, garage) become `None`, genuine gaps are imputed (for example lot frontage by neighborhood median), two known outliers are removed, and new features (`TotalSF`, `HouseAge`, `PricePerSqft`) are engineered. The result is 1,458 homes with no missing values. Eight Matplotlib and Seaborn charts cover price distribution, location comparison, area vs price, quality, trend, correlations, predicted vs actual and feature importance.



2. Machine-learning price model

Three models are compared on a held-out 20% test set: Linear Regression, Decision Tree and Random Forest. The target is `log(SalePrice)`. The neighborhood is **target-encoded** inside the pipeline (a plain one-hot encoding was almost ignored by the trees, which made neighborhood changes have no effect on predictions). The Random Forest is saved with `joblib` and served by the API.

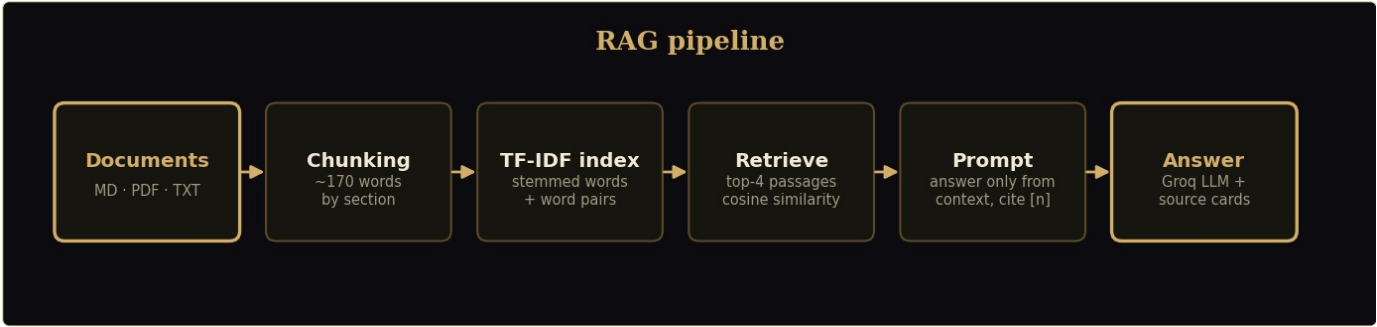
3. Recommendation system

Candidates are filtered by budget, bedrooms and neighborhoods, then scored on **budget fit**, **value against the AI fair-value estimate**, build quality and preference matches (garage, newer build, quality, lot, size, basement). Each result carries plain-language reasons, for example *"Listed 9% below the AI fair-value estimate of \$268,289"*.

4. Generative AI assistant

A structured system prompt defines the role, injects real market facts from the data, forbids invented statistics, requires a balanced view of risks, adds legal and financial disclaimers and caps answer length. Property summaries use a separate, data-only prompt.

5. RAG knowledge assistant

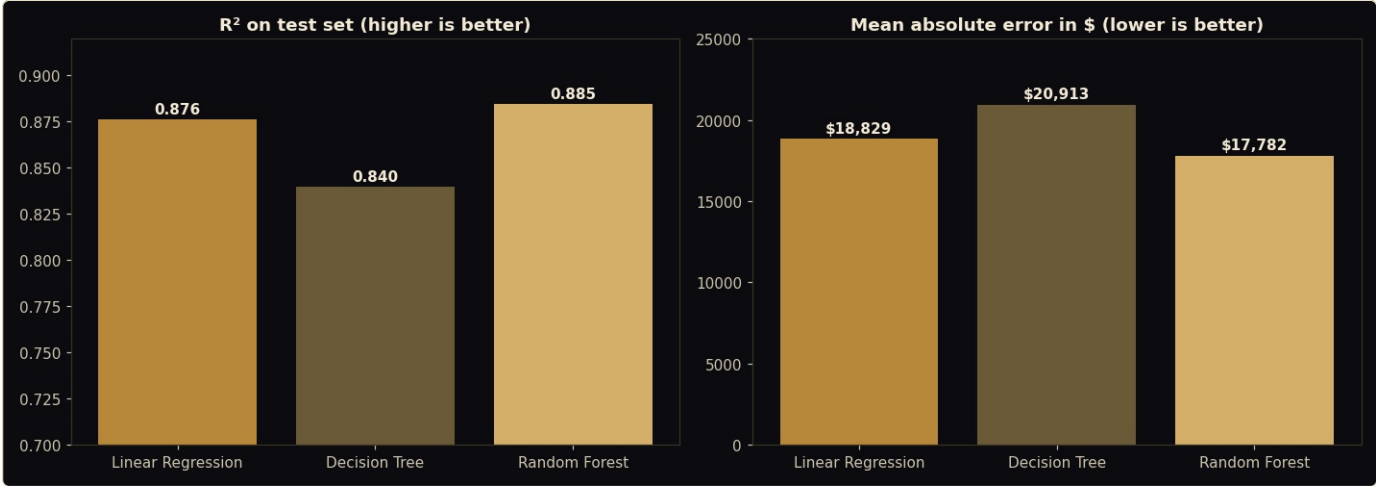


Documents are split by section into roughly 170-word passages and indexed with TF-IDF over lightly stemmed words and word pairs, with section titles weighted double. A question retrieves the top four passages; the LLM must answer **only** from them and cite `[n]`. If nothing relevant is found, the assistant says so instead of guessing. The knowledge base contains four documents: *Property Buying Guide*, *Registry and Legal Process* (Iowa), *Investment Tips* (with a worked cap-rate example) and *Ames Area Information* (generated from the dataset).

Model results

Evaluated on 292 held-out homes:

Model	R ²	MAE	RMSE	5-fold CV R ²
Linear Regression	0.876	\$18,829	\$26,144	0.859
Decision Tree	0.840	\$20,913	\$29,761	0.769
Random Forest (deployed)	0.885	\$17,782	\$25,251	0.847



- **RAG retrieval:** on a 15-question evaluation set (`python ml/eval_retrieval.py`) the right document is ranked first for 15 of 15 questions (100%); a plain TF-IDF baseline reached 14 of 15 (93%).
- **63%** of predictions are within 10% of the real sale price, **77%** within 15% and **87%** within 20%. The median error is **6.6%**.
- The Random Forest has the best test R² and lowest error. Linear Regression has a slightly higher cross-validation R², so the margin between the two is small; the Random Forest was chosen for its lower error and its ability to capture non-linear effects such as the premium on high build quality.

Tech stack

Layer	Tools
Data and ML	Python, Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, joblib
Backend	FastAPI, Pydantic, Jinja2, ReportLab, pypdf, httpx
GenAI and RAG	Groq API (Llama 3.3 70B), prompt engineering, TF-IDF retrieval
Frontend	HTML, CSS, vanilla JavaScript (no framework), Google Fonts
DevOps	GitHub, Vercel, Firebase Hosting

Getting started

Prerequisites

Python 3.10 or newer and Git. A free [Groq API key](#) is optional.

Install and run

```
git clone https://github.com/<your-username>/propmind.git
cd propmind

python -m venv .venv
.venv\Scripts\activate          # Windows   (macOS/Linux: source .venv/bin/activate)
pip install -r requirements-dev.txt

python ml/train.py              # optional: retrain with your library versions
copy .env.example .env         # macOS/Linux: cp .env.example .env
uvicorn app.main:app --reload
```

Open **http://127.0.0.1:8000**. Interactive API docs are at **/docs**.

Environment variables

Variable	Required	Description
GROQ_API_KEY	No	Enables full AI answers. Without it the assistant runs in offline mode
GROQ_MODEL	No	Defaults to llama-3.3-70b-versatile

Notebook and tests

```
jupyter notebook notebooks/PropMind_Analysis.ipynb    # or open it in VS Code and Run All
python tests/test_core.py                             # or: python -m pytest -q
```

The saved notebook already contains every output and chart.

API reference

Method	Endpoint	Purpose
GET	/api/health	Status, model name and AI mode
GET	/api/meta	Neighborhoods, preference keys, model metrics
POST	/api/predict	Price estimate, range, confidence, what-if
POST	/api/recommend	Ranked homes with reasons
GET	/api/properties?ids=1,2	Property details by id
POST	/api/summary	AI summary of one property
POST	/api/chat	General AI advisor
POST	/api/rag/ask	Answer from documents, with sources
GET	/api/rag/docs	List indexed documents
POST	/api/rag/upload	Upload a PDF, TXT or MD document
GET	/api/insights	Neighborhood statistics
POST	/api/report	Download a PDF valuation report

Example:

```
curl -X POST http://127.0.0.1:8000/api/predict \
-H "Content-Type: application/json" \
-d
'{"GrLivArea":1710,"BedroomAbvGr":3,"FullBath":2,"OverallQual":7,"YearBuilt":2003,"GarageCars":2,"TotalBsmntSF":856,"LotArea":8450,"Neighborhood":"CollgCr"}'
```

Deployment

The backend and website run together on **Vercel**. The same website can also be hosted on **Firebase Hosting** as a static frontend that calls the Vercel API. Neither free plan needs a billing card.

1. GitHub

```
git init && git add .
git commit -m "PropMind: AI Real Estate Intelligence Platform"
git branch -M main
git remote add origin https://github.com/<your-username>/propmind.git
git push -u origin main
```

2. Vercel (backend and website)

- 1. On **vercel.com** choose **Add New → Project** and import the repository.
- 2. Set the framework preset to **Other** and leave the build settings empty.
- 3. Add the environment variable `GROQ_API_KEY`.
- 4. Deploy. The site and API are live at `https://<project>.vercel.app`.

`vercel.json`, `index.py` and `.vercelignore` are already configured. Vercel functions are stateless, so documents uploaded from the web page can disappear between requests; the four built-in documents always remain. If the bundle exceeds Vercel's size limit, deploy the same repository on Render or Hugging Face

Spaces.

3. Firebase Hosting (static frontend)

```
npm install -g firebase-tools
firebase login
copy .firebaserc.example .firebaserc      # put your Firebase project id inside
python ml/build_static.py --api https://<project>.vercel.app
firebase deploy --only hosting
```

Project structure

```
propmind/
├── app/
│   ├── main.py           FastAPI app: pages and JSON API
│   ├── services/         predictor, recommender, rag, assistant, llm, prompts
│   ├── templates/        base + 7 pages (Jinja2)
│   └── static/            css/, js/, charts/
├── data/                 train.csv (Kaggle) and properties_clean.csv
├── docs/                  4 RAG documents, images/, screenshots/
├── ml/                    train.py, make_area_doc.py, make_report_figures.py,
│                           build_notebook.py, build_static.py
├── models/                price_model.joblib, model_stats.json, final_eval.json
├── notebooks/             PropMind_Analysis.ipynb
├── reports/               report (DOCX+PDF), presentation (PPTX+PDF), README (PDF)
├── tests/test_core.py
├── index.py, vercel.json  Vercel endpoint
├── firebase.json          Firebase Hosting config
└── requirements.txt, requirements-dev.txt, .env.example
```

Troubleshooting

Problem	Fix
ModuleNotFoundError	Activate the virtual environment and run <code>pip install -r requirements-dev.txt</code>
Model loading error or version warning	Run <code>python ml/train.py</code> to rebuild the model with your installed scikit-learn
Chatbot says "Offline mode"	Create <code>.env</code> from <code>.env.example</code> , add <code>GROQ_API_KEY</code> , restart the server
Pages load but show "Is the API running?"	Open the site through <code>http://127.0.0.1:8000</code> , not by double-clicking an HTML file
Port 8000 already in use	Run <code>uvicorn app.main:app --reload --port 8001</code>
Uploaded document disappears (on Vercel)	Expected on stateless hosting; add it again or keep it in <code>docs/</code>

Limitations and roadmap

Limitations. The model learns from one city and the years 2006-2010 and uses nine inputs, so estimates are educational, not appraisals. The legal and investment documents are general educational material, not legal, tax or financial advice. Demo accounts live only in the visitor's browser. TF-IDF retrieval matches words, not meaning.

Roadmap. - Embedding-based retrieval with a vector database for semantic search - Real accounts and saved data with Firebase Authentication and Firestore - Gradient boosting models and a larger, more recent dataset - Map view with listing photos and an image-based condition score - Multilingual interface (English and Urdu)

Dataset, acknowledgments and license

- **Dataset:** [House Prices: Advanced Regression Techniques](#) (Kaggle, MIT license), based on the Ames Housing data compiled by Dean De Cock.
- **Built with** FastAPI, scikit-learn, Groq and the open-source Python ecosystem.
- **License:** MIT. See [LICENSE](#).

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